Sieve of Eratosthenes (1–100)

Mathematical description. Fix $n = 100$. Let $S = \{2, 3, \ldots, n\}$. For each integer p defined as the smallest element of S not yet processed, declare p *prime* and remove from S every integer of the form kp with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of S that remain after this process are precisely the prime numbers $\leq n$.

Algorithm (short).

1. Initialize the list $2, 3, \ldots, n$.
2. For each current smallest unprocessed p (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \ldots$ up to n.
3. Stop once $p^2 > n$ (i.e. $p > \lfloor \sqrt{n} \rfloor$). Remaining numbers are primes.

Note. 1 is neither prime nor composite.

	2	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

Primes The primes ≤ 100 are:

2, **3**, **5**, **7**, **11**, **13**, **17**, **19**, **23**, **29**, **31**, **37**, **41**, **43**, **47**, **53**, **59**, **61**, **67**, **71**, **73**, **79**, **83**, **89**, **97**.

L1: Divisibility

- Elementary number theory is the study of divisibility properties of the integers ($\mathbb{Z}$).
- This topic dates back to 300 B.C, and many ideas in this class (and used widely today) appeared in "Elements" by Euclid i.e. Enclid's proof of the infinitude of primes, dividion (Enclidean) algorithm, perfect numbers, Pythagorean triples etc.
  Definition: Let $a, b \in \mathbb{Z}$. Then a divides b, denoted $a \mid b$, if there exists $c \in \mathbb{Z}$ such that $b = ac$. If $a \mid b$, then a is said to be a divisor or factor of b. The notation $a \times b$ means that a does not divide b.

Example: 316 since there exists $c \in \mathbb{Z}$ such that $6 = 3c$. In particular, $c = 2$.

- $31 - 6$ since there exists $c \in \mathbb{Z}$ such that
  $-6 = 3c$. In particular, $c = -2$.
- If $a \in \mathbb{Z}$, then $a10$ since there exists
  $c \in \mathbb{Z}$ such that $0 = ac$. In particular, $c = 0$. Any integer divides 0 .
- If $a \in \mathbb{Z}$ and $O \mid a$, then there exists $c \in \mathbb{Z}$ such that $a = Oc$ if and only if $a = 0$.
  Remark: The notation alb does not have the same meaning as a/b and b/a. The statement alb is about the relationship between two integers i.e. a divides into b with no remainder.
  The notations a/b and b/a mean $a \div b$ and $b \div a$ respectively, and each are rational numbers. Note that alb implies that a divides b exactly b/a times provided that $a \neq 0$.
  Remark: We say O1O because it is consistent with our definition of divisibility. It should not be interpreted as implying that 0 divides O exactly "O/" times. The form "O/O"
  is indeterminate, and has no meaning.
  Proposition 1.1: Let $a, b, c \in \mathbb{Z}$. If $a \| b$ and $b \mid c$, then alc.
  Proof: Since alb and blc, there exist $e, f \in \mathbb{Z}$ such that $b = ae$ and $c = bf$. Then

$$c = bf = (ae)f = a(ef),$$

and so $a \mid c$.

- The previous Proposition proves that divisibility is transitive.
  Proposition 1.2: Let $a, b, c, m, n \in \mathbb{Z}$. If cla, and clb, then $clma + nb$.
  Proof: Since $c \mid a$ and $c \mid b$, there exist e, $f \in \mathbb{Z}$ such that $a = ce$ and $b = cf$. Then

$$ma + nb = mce + ncf = c(me + nf),$$

and so c lma+nb.
Definition: The expression $ma + nb$ is said to be an integral linear combination of a and b.

Definition: Let $x \in \mathbb{R}$. The greatest integer function of x, denoted $[x]$, is the greatest integer less than or equal to x.

- The existence of a greatest integer less than or equal to a given real number follows from
  (1) Well ordering principle of $\mathbb{Z}$: any non-empty set of positive integers contains a least element.
  (2) Archimedean property of $\mathbb{R}$.

Example: If $a \in \mathbb{Z}$, then $[a] = a$.
The converse that if $[a] = a$ for some $a \in \mathbb{R}$, then $a \in \mathbb{Z}$ is also true.

- $[5/3] = 1, [\pi] = 3, [-5/3] = -2, [-\pi] = -4$.

Lemma 1.3: Let $x \in \mathbb{R}$. Then $x - 1 < [x] \leqslant x$.
Proof: Since the greatest integer function of x is less than or equal to x, the second inequality is clear. For the first inequality, assume that $x - 1 \geqslant [x]$. Then $[x] + 1 \leqslant x$, and

$$[x] < [x] + 1 \leqslant x.$$

Since $[x]+1 \in \mathbb{Z}$, this contradids the fact that $[x]$ is the grentest integer less than or equal to x.

Hence $x-1<[x]$, as required.

- When one integer (the divisor) is divided into another integer (the dividend), to obtain an integer quotient, an integer remainder is obtained.
  Example: $\quad 5=3 \cdot 1+2$
- "The dividend is equal divisor $\frac{-3}{2}$ to the divisor times the Guotient plus the remainden" remainder
- The remainder is strictly less than the divisor.

Theorem 1.4 (The Division Algorithm):
Let $a, b \in \mathbb{Z}$ with $b>0$. Then there exist unique $q, r \in \mathbb{Z}$ such that

$$
\begin{aligned}
& a=bq+r, 0 \leq r<b . \\
& \ell, 0 \text { remainder } \\
& \text { quotient }
\end{aligned}
$$

Proof: Let $q:=\left[\frac{a}{b}\right]$ and $r=a-b\left[\frac{a}{b}\right]$.
Then $a=bq+r$. We now prove that $0 \leqslant r<b$.

By Lemma 1.3 we have that

$$
\frac{a}{b}-1<\left[\frac{a}{b}\right] \leqslant \frac{a}{b} .
$$

Multiplying all terms of this inequality by $-b$ gives

$$
b-a>-b\left[\frac{a}{b}\right] \geqslant-a .
$$

Reversing the inequality and adding a to all terms gives

$$
0 \leqslant a-b\left[\frac{a}{b}\right]<b,
$$

which is exactly $0 \leqslant r<b$, as desired. Thus the q and r defined above have the desired properties. It remains to show the uniqueness of q and r. Assume that

$$
\begin{aligned}
& a=bq_1+r_1, \quad 0 \leqslant r_1<b . \\
& a=bq_2+r_2, \quad 0 \leqslant r_2<b .
\end{aligned}
$$

We have $0=a-a=bq_1+r_1-\left(bq_2+r_2\right)$

$$
=b\left(q_1-q_2\right)+\left(r_1-r_2\right),
$$

which implies that

$$
r_2-r_1=b\left(q_1-q_2\right)
$$

Hence $b \mid r_2-r_1$. Now $O \leqslant r_1<b$ and
$0 \leqslant r_2<b$ imply that $-b<r_2-r_1<b$.
Thus we have $r_2-r_1=0$ i.e. $r_1=r_2$. Thus (1)
becomes

$$0 = b\left(q_1 - q_2\right).$$

Since $b \neq 0$ we have that $q_1 - q_2 = 0$
i.e. $q_1 = q_2$. Thus we have the uniqueness claim for q and r.

- Observe that $r = 0$ in the Division Algorithm if and only if b/a.
  Example: Suppose $a = -5$ and $b = 3$.
  we have $q = \left[\frac{a}{b}\right] = \left[-\frac{5}{3}\right] = -2$,
  and $\quad r = a - b\left[\frac{a}{b}\right] = -5 - 3(-2) = 1$.
  Check:

$$\begin{aligned} a &= \log + r \\ -5 &= 3(-2) + 1 \end{aligned}, 0 \leqslant r = 1 < 3.$$

Remark: What is " 5 divided by 3?" Someone might respond with "-1 with remainder - 2 "
since $-\frac{5}{3} = -1 - \frac{2}{3}$. Although $q = -1$
and $r = -2$ satisfy $a = bq + r$, the condition that $0 \leqslant r < b$ in Theorem 1.4 is no longer true. Hence, division in the context of the Division algorithm must be performed carefully so that $a = bq + r$ and $0 \leq r < b$ both hold.

Definition: Let $n \in \mathbb{Z}$. Then n is said to be even if $2 \ln$ and n is said to be odd if $2 \times n$.

Example: The set of even integers is

$$\{\cdots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$

The set of odd integers is

$$\{\cdots, -7, -5, -3, -7, 0, 1, 3, 5, 7, \cdots\}$$

L2: Prime Numbers

Definition: Let $p \in \mathbb{Z}$ with $p > 1$. Then p is said to be a prime if the only positive divisors of p are 1 and p. If $n \in \mathbb{Z}, n > 1$, and n is not prime, then n is said to be composite.

- The positive integer 1 is neither prime non composite. We will give a compelling reason for this later in the course.
  Example: The prime numbers between 2 and 50 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43$, and 47. All primes except 2 are odd.
  Lemma 1.5: Every integer greater than 1 has a prime divisor.
  Proof: Assume there exists an integer $n > 1$ that has no prime divisor. By the well ordering principle we may assume that n is the least such integer. Clearly $n \mid n$, so n is not prime. Thus n is Composite and Consequently there exist $a, b \in \mathbb{Z}$ such that $n = ab$, where $1 < a < n$ and $1 < b < n$. Since $1 < a < n$, we have that a has a prime divisor, say p, so that pla. But aln, so we have that pln by Proposition 1.1 (transitivity of divisibility). Contradiction! Thus every integer greater than I has a prime divisor.
  Theorem 1.6 (Enclid) There are infinitely many prime numbers.
  Proof: Assume that there are only finitely many prime numbers, say $P_1, P_2, \ldots, P_n$. Consider the integer $N := P_1 P_2 \ldots P_n + I$. By Lemma 1.5 we know that N has a prime divisor, say ρ. Thus $p = p_i$ for some i, $i = 1, \cdots, n$. Then $p \mid (N - p_1 p_2 \cdots p_n)$ by Proposition 1.2, so P$|$1, a contradiction. Thus there are infinitely many prime numbers.

Proposition 1.7: Let n be a composite number. Then n has a prime dividor p with $p \leq \sqrt{n}$. Proof: Since n is composite, there exist $a, b \in \mathbb{Z}$ such that $n = ab, 1 < a < n, 1 < b < n$, and without loss of generality, $a \leq b$. We must have $a \leqslant \sqrt{n}$, otherwise if $a > \sqrt{n}$ we have $n = ab > \sqrt{n}\sqrt{n} = n$, which is not possible.
By Lemma 1.5 we know that a must have a prime divisor, say p. Since pla and aln we have pln by Proposition 1.1 (transitivity of divisibility). Furthermone, $p \leq a \leq \sqrt{n}$, so p is the desired prime factor of n.

- Proposition 1.7 gives a method for finding all the prime numbers less than or equal to a specified integer $n > 1$ This is called the sieve of Eratosthenes.
  Example: Suppose one wants to find all the primes less than 50. By Proposition 1.7
  any composite number less than 50 must have a prime dividor less than or equal to $\sqrt{50} \approx 7.07$ i.e. $2, 3, 5, 7$. Deleting all multiples of these primes (except for themselves) gives:
  11 12 13 14 18 16 17 18 19 26
  21 22 23 24 25 26 27 28 29 30
  31 32 35 34 35 36 37 38 39 46
  41 42 43 44 45 46 47 48 49 50
  Remaining numbers are prime:
  $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41,$
  $43, 47.$

- Proposition 1.7 gives a very crude primality test i.e. an algorithm to test whether a given $n > 1$ is prime or composite i.e. Check whether n is divisible by any prime number $p \leq \sqrt{n}$. If n is divisible by such a prime, then n is composite. Otherwise, n is

prime.

- We will discuss more efficient primality tests later in the course.
- If one does larger experiments with the sieve of Eratos thenes one might guess that the primes are more sparsely distributed as you progress through larger and larger integers. The next Proposition shows that there exist arbitrarily long sequences of Consecutive positive integers that don't Contain Prime numbers
  Proposition 1.8: For any positive integer n, there are at least n consecutive composite positive integers.
  Proof: Given $n \geqslant 1$, consider the sequence

$$(n+1)! + 2, (n+1)! + 3, \cdots, (n+1)! + n + 1.$$

Let i be an integer such that $2 \leq i \leq n+1$. Since $i|(n+1)|$ and $i \mid i$, we have $i|(n+1)| + i$ for $2 \leq i \leq n+1$, by Proposition 1.2. Thus each integer in the sequence is composite.

- It also seems that one can find primes (6) very close to one another
  i.e. $(3,5),(5,7),(11,13), \ldots$
  A pair ($P, P+2$) with both entries prime are called twin primes.
  Conjecture: There are infinitely many prime numbers P for which $P+2$ is also Prime.

- There has been very recent breakthroughs on this problem. Let $p_1, p_2, \ldots$ denote the sequence of primes. Zhang (2013) proved that there are infinitely many primes p_n such that $P_{n+1}-P_n \leqslant 70000000$. This has been improved by others, Maynard, Tao, $\cdots$, but not all the way to 2 .
  One of the most famous (and classical) results about the Primes is the Prime Number Theorem (PNT). This was conjectured by Gauss in 1793 and was proved independently
  in 1896 by C.J. de la Vallée Poussin 7 and J. Hadamard. The PNT gives an estimate for the number of primes up to a real positive number x.
  Definition: Let $x \in \mathbb{R}$ with $x>0$. Then

$$\pi(x):=\mid\{p: p \text{ prime; } \mid<p \leqslant x\} \mid .$$

Theorem 1.9 (PNT): $\lim_{x \rightarrow \infty} \frac{\pi(x)}{x / \log x}=I$.

- Theorem 1.9 is equivalent to saying that $\pi(x)$ may be approximated by $\frac{x}{\log x}$ as $x \rightarrow \infty$.
- The proof of PNT is beyond the scope of this class (requires Complex analysis and more advanced number theory).
  Many unsolved problems in number theory deal with integers that are expressible in certain forms.

Conjecture (Goldbach, 1742):

Every even integer greater than 2 can be expressed as the sum of two (not necessarily distunct) prime numbers.
i.e. $4=2+2, 6=3+3, 8=3+5$, and so on

- Goldbach's conjecture has been Verified experimentally up to some large finite point, but a proof still eludes the research community. In fuct, Hardy and Littlewood predicted a frmula for the number of representations an even integer has in terms of a sum of two primes. Still an open problem!
- We now consider other families of Primes and close with some more open conjectures.
  Definition: Any prime number expressible in the form 2^p-1 with p prime is said to be a Mersenne prime.

Example: The first five Mersenne primes
are $3=2^2-1, 7=2^3-1, 31=2^5-1$,
$127=2^7-1$ and $8191=2^{13}-1$.
Note that $2''-1=2047=(23)(89)$ is not a Mersenne prime.
Conjecture: There are infinitely many Mersenne primes.

Another interesting family of prime numbers are due to Fermat.
Definition: Any prime number expressible in the form $2^{2^n}+1$ with $n \in \mathbb{Z}$ and $n \geqslant 0$ is said to be a Fermat Prime.
Example: The first five Fermat primes are $3=2^{2^0}+1, 5=2^{2^1}+1, 17=2^{2^2}+1$,
$257=2^{2^3}+1$, and $65537=2^{2^4}+1$.

- Fermat in 1640 conjectured that any.
  number expressible in the form $2^{2^n}+1$
  with $n \in \mathbb{Z}$ and $n \geqslant 0$ is prime.
- Fermat's conjecture was disproved in 1732 by Enler who proved that $641 / 2^{2^5}+1$ i.e. $2^{2^5}+1$ is not a Fermat prime.

Conjecture: There are exactly five Fermat primes.

- The last family we consider is due to Hardy and Littlewood 1922 :
  Conjecture: There are infinitely many primes expressible in the form n^2+1 where n is a positive integer.
  Remark: In view of the above conjecture it may be tempting to conjecture there are infinitely many primes of the form n^2-1 for n a positive integer. However,

$$n^2 - 1 = (n + 1)(n - 1),$$

which rules out the primality of $n^2 - 1$ for $n > 2$. Factorisations can be a powerful (II) tool.

L3: Greatest Common divisors

- Given $a, b \in \mathbb{Z}$, both not zero, consider

$$\delta := \{c : c \in \mathbb{Z}, c \mid a, \text{ and } c \mid b\}.$$

The set δ is non-empty since $\pm 1 \in S$, and finite since O is the only integer that has an infinite number of divisors, and at least one of a and b are non-zero. Thus it makes sense to speak of the greatest element of $S_,$ and that element is necessarily positive.
Definition: Let $a, b \in \mathbb{Z}$ with a and b not both zero. The greatest common divisor of a and b, denoted (a, b), is the greatest positive integer d such that dla and dib. If $(a, b) = 1$, then a and b are said to be relatively prime.
Remark: Note that $(0, 0)$ is undefined.
Remark: If $(a, b) = d$, then

$$(-a, b) = (a, -b) = (-a, -b) = d. \quad (*)$$

Thus without loss of generality we vestrict our
computations of the gcd of two non-negative integers in the following example.
Example:

- Divisors of $24 : \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12, \pm 24$

Divisors of $60 : \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 10,$

$$\pm (2), \pm 15, \pm 20, \pm 30, \pm 60$$

Thus $(24, 60) = 12$.

- In general, if $a \in \mathbb{Z}$ with $a \neq 0$, then $(a, 0) = |a|$. $\quad (**)$
- Divisors of $35 : \pm 1, \pm 5, \pm 7, \pm 35$.

Thus $(24, 35) = 1$, and 24 and 35 are velatively prime.
Remark: If $(a, b) = d$, then $d \mid a$ and $d \mid b$.
Proposition 1.10: Let $a, b \in \mathbb{Z}$ with $(a, b) = d$. Then $(a/d, b/d) = 1$.
Proof: Let $d' := (a/d, b/d)$. Then $d' \mid a/d$ and $d' \mid b/d$, so there exist $e, f \in \mathbb{Z}$ such
that $a/d = d'e$ and $b/d = d'f$. Thus $a = d'de$ and $b = d'df$, and consequently we have $d'd \mid a$ and $d'd \mid b$. This implies that $d'd$ is a common divisor of a and b, and since d is the gcd of a and b, we have $d' = 1$, as required.

Proposition 1.11: Let $a, b \in \mathbb{Z}$ with a and b not both zero. Then
$(a, b) = \min\{ma + nb : m, n \in \mathbb{Z}, ma + nb > 0\}$.
Proof: We first need to prove that the minimum exists. Observe that

$$\{ma + nb : m, n \in \mathbb{Z} \partial ma + nb > 0\} \neq \varnothing,$$

Since, without loss of generality, $a \neq 0$, and then either $la + Ob > 0$ or $- \mid a + Ob > 0$. Thus min $\{ma + nb : m, n \in \mathbb{Z}, ma + nb > 0\}$ exists by the well-ordering principle, denote it d. Then $d = m'a + n'b$ for some $m'n' \in \mathbb{Z}$.

We first prove that dla and d/b. By the (4) division algorithm there exist $q, r \in \mathbb{Z}$ such that

$$a = dq + r, \quad 0 \leqslant r < d$$

Now,

$$
\begin{aligned}
r = a - dq &= a - (m'a + n'b)\, q \\
&= (1 - qm')\, a - qn'b_J
\end{aligned}
$$

and we have that n is an integral linear combination of a and b. Since $O \leq r < d$ and d is the minimum positive integral linear combination of a and b, we have $r = 0$, from which $a = dq + r$ implies that $a = dq$. Thus dla, and by a similar argument, $d \mid b$. It remains to show that d is the greatest common divisor. Let c be any common divisor of a and b, then cla and clb. Then $c\,(m'a + n'b = d$ by Proposition 1.2, and so $c \leqslant d$.

- Proposition 1.11 tells us that $(a, b) = d$ implies that d may be expressed as an integral linear combination of a and b. The converse of this statement is true if $d = 1$, namely, if 1 is expressible as an integral linear combination of a and b, then $(a, b) = 1$. Why?
  Well, $1 = ma + nb$ for some $m, n \in \mathbb{Z}$, and any $d \in \mathbb{Z}$ such that $d \mid a$ and $d \mid b$ must then satisfy $d \mid ma + nb = 1$ by Proposition (.2, so $(a, b) = 1$.

- Proposition 1.11 also can be used to show that a common divisor is not only less them the greatest common diviser, but also divides the greatest common divisor:
  $(a, b) = am + bn$ for some $m, n \in \mathbb{Z}$, so if
  dla and $d \mid b$, then $d \mid am + bn = (a, b)$ by Proposition 1.2.

Definition: Let $a_1, a_2, \ldots, a_n \in \mathbb{Z}$ with $a_1, a_2, \ldots, a_n$ not all zero. The greatest common divitor of $a_1, a_2, \ldots, a_n$, denoted $(a_1, a_2, \ldots, a_n)$ is the greatest integer d such that $d \mid a_i$ for all $i = 1, \ldots, n$. If $(a_1, a_2, \ldots, a_n) = I$, then $a_1, a_2, \ldots, a_n$, are said to be velatively prime. If $(a_i, a_j) = 1$ for all pairs i, j with $i \neq j$, then $a_1, a_2, \ldots, a_n$ are said to be pairwise relatively prime.
Example: The common dividoss of $24, 60, 30$ are $\pm 1, \pm 2, \pm 3, \pm 6$. Thus $(24, 60, 30) = 6$.

Pairwise relatively prime integers are relatively prine, but the converse is not always true.
Example: Check that $(24, 60, 49) = 1$, so $24, 66, 49$ are relatively prime. However, $(24, 60) = 12 \neq 1$, so they are not pairwise

relatively prime.

The Enclidean Algorithm

- We want a move efficient method for computing the god of two integers than just listing of their divisors.
  Lemma 1.12: If $a, b \in \mathbb{Z}, a \geqslant b > 0$, and $a = bq + r$ with $q, r \in \mathbb{Z}$, then $(a, b) = (b, r)$.
  Proof: Let c be a common divisor of a and b. Then c/a and c/b imply that $c \mid a - qb = r$, and so c is a common divisor of b and r.

Now let c be a common divisor of b and r, so that $c \mid b$ and $c \mid r$. Then $c \mid qb + r = a$ by Proposition 1.2 , from which $c(a)$ and so c is a common divisor of a and b.
The set of common divisors of a and b is the same as the set of common divisors of b and r, and so $(a, b) = (b, r)$.

- The Enclidean algorithm for computing
  the ged of two positive integers comes from Book VII of Enclid's "Elements".
  Theorem 1.13: Let $a, b \in \mathbb{Z}$ with $a \geqslant b > 0$. By the division algorithm (Theorem 1.4), there exist $q_1) r_1 \in \mathbb{Z}$ such that

$$a = bq_1 + r_1, 0 \leqslant r_1 < b.$$

If $r_1 > 0$, there exist $q_2, r_2 \in \mathbb{Z}$ by the division algosithm such that

$$b = r_1 q_2 + r_2, \quad 0 \leqslant r_2 < r_1.$$

If $r_2 > 0$, there exist $q_3, r_3 \in \mathbb{Z}$ such that

$$r_1 = r_2 q_3 + r_3, \quad 0 \leq r_3 < r_2$$

Continue this process. Then $r_n = 0$ for some $n \geqslant 1$, and

$$(a,b) = \begin{cases} b & \text{if } n = 1 \\ r_{n-1} & \text{if } n > 1 \end{cases}$$

Proof: Note that $r_1 > r_2 > r_3 > \dots$ If $r_n \neq 0$ for all $n \geqslant 1$, then $r_1, r_2, r_3, \dots$ is an infinite, strictly decreasing sequence of positive integers, which is not possible!
Thus $r_n = 0$ for some n. If $n > 1$ we repeatedly apply Lemma 1.12 to obtain
$(a,b) = (b, r_1) = (r_1, r_2) = (r_2, r_3) = \cdots = (r_{n-1}, r_n)$

$$= (r_{n-1}, 0) = r_{n-1})$$

as desired. If $n = 1$, the statement is clear.

Remark: The Enclidean Algosithm finds the ged of a, b when they are positive integers. By (*) and (**), all other cases are reducible to this case or trivial.
Example: Find $(803, 154)$ using the Euclidean algorithm:

$$\begin{align} 803 &= 154 \cdot 5 + 33 \tag{1} \\ 154 &= 33 \cdot 4 + 22 \tag{2} \\ 33 &= 22 \cdot 1 + 11 \tag{3} \\ 22 &= 11 \cdot 2 + 0 \tag{4} \end{align}$$

Thus $(803, 154) = 11$.

- Recall that Rroposition 1.11 says that the ged of two integen is expressible as an integral linear combination of the two integers. The Euclidean algorithm gives a Systematic way to find Such a linear combination.
  Example: Express $(803, 154)$ as an integral linear combination of 803 and 154.
  We work through the previous example backward. We have

$$\begin{align} (803,154) = 11 &= 33 - 22 \text{ by} \\ &= 33 - (154 - 33 \cdot 4) \text{ by } (2) \\ &= 33 \cdot 5 - 154 \\ &= (803 - 154 \cdot 5)5 - 154 \text{ by } (0) \tag{0} \\ &= 803 \cdot 5 - 154 \cdot 26 \\ &= 5.803 + (-26) \cdot 154. \end{align}$$

Remark: This integral linear combination of
$(803, 154)$ is not unique. Can verify that $11 = 19.803 - 99.154$ is another such expression. We will study this more later in the course.

- No integral linear combination of 803 and

154 can yield any of the integers $1, 2, 3, \dots, 10$ by Proposition 1.11.

L4: The Fundamental Theorem of Arithmetic

Lemma 1.14 (Euclid): Let $a, b, p \in \mathbb{Z}$ with p prime. If $plab$, then pla or plb.
Proof: Assume that $p + a$. Then $(a, p) = 1$. We must show that plb. By Proposition (.11, there exist $m, n \in \mathbb{Z}$ such that $ma + np = 1$. Also, plab implies that $ab = pc$ for some $c \in \mathbb{Z}$. After multiplying both sides of $ma + np = I$ by b, we have that $mab + npb = b$. Then $ab = pc$ implies that $mpc + npb = b$, or equivalently $p(mc + nb) = b$. Thus we have $p \mid b$, as desired.

Remark: Note that Lemma 1.14 does not hold if p is composite i.e. take $p = 6a = 2, b = 3$.

Corollary 1.15: Let $a_1, a_2, \ldots, a_n, p \in \mathbb{Z}$ with p Prime. If $p\left(a_1 a_2 \cdots a_n\right)$ then $p\left(a_i\right.$ for some i.

Proof: We use induction on n. The statement for $n = 1$ is trivially true. The statement for $n = 2$ is Lemma 1.14. Assume that $k \geqslant 2$ and that the desired statement is true for $n = k$ i.e. that $p \mid a_1 \ldots a_k$ implies that plai for some i with $1 \leqslant i \leqslant k$. Now we need to prove that $p \mid a_1 \ldots a_{k+1}$ implies that $p \mid a_i$ for some i with $1 \leq i \leq k+1$, So that the desired statement holds for $n = k+1$. Observe that $p \mid a_1 \ldots a_{k+1}$ implies that $p \mid \left(a_1 \ldots a_k\right) a_{k+1}$, so Lemma 1.14 then implies that $p \mid a_1 \ldots a_k$ or $p \mid a_{k+1}$.
If $p \mid a_{k+1}$ we are done. If $p \nmid a_{k+1}$, then
$p / a_1 \ldots a_k$ which implies that p / a_i for some
i with $1 \leq i \leq k$ by the induction hypothesis.
The desired statement holds for $n = k+1$,
as required.

Theorem 1.1.6(FTA): Every integer greater than I can be expressed in the form $p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}$ with $P_1, P_2, \cdots, P_n$ distinct primes and $a_1, a_2, \ldots, a_n$ positive integers. This form is said to be the prime factorisation of n. The prime factorisation is unique except up to permutation of the $p_i a_i$.
Remark: One of the main reasons 1 is not a prime is because the uniqueness of prime factorisations would be lost otherwise.
Proof (of FTA): Assume, by way of contradiction, that $k > I$ is an integer that does not have a prime factorisation. By the well-ordering principle we may assume that k is the least such integer. Note that k can't be prime, otherwise it would be of the desired form. Thus k is composite and $k = ab$ with $| < a < k$ and $| < b < k$. Due to the minimality of k, both a and b have
prime factorisations, and so to does $k = ab$. Contradiction! Thus every integer greater than I has a prime factorisation.
We now need to prove the uniqueness of prime factorisation (up to permutations). Assume k has two prime factorisations, say

$$k = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n} = q_1^{b_1} q_2^{b_2} \cdots q_m^{b_m},$$

where $p_1, p_2, \cdots, p_n$ are distinct primes,
$q_1, q_2, \ldots, q_m$ are distinct primes, and $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_m$ are positive integers. Without loss of generality, we may assume that $p_1 < p_2 < \ldots < p_n$ and $q_1 < q_2 < \cdots < q_m$. We must show that $\begin{cases} n = m \\ p_i = q_i, i = 1, \ldots, n \\ a_i = b_i, i = 1, \ldots, n. \end{cases}$

Given a P_i, we have
$p_i \mid q_1^{b_1} q_2^{b_2} \ldots q_m^{b_m}$,
which implies that $p_i \mid q_j$ for some j by Corollary 1.15. Thus $P_i = q_j$ for some j. Similarly, given some q_k, we have $q_k = P_e$ for some l. Thus $n = m$. By the ordering of the p_i 's and q_j s we have $p_i = q_i$ for $i = 1, \ldots, n$. Consequently,

$$k = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n} = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}.$$

Assume, by way of contradiction, that $a_i \neq b_i$ for some i. Without loss of
generality we may assume that $a_i < b_i$. Then $p_i^{b_i} \mid p_1^{a_1} \ldots p_n^{a_n}$,
which implies that

$$p^{b_i - a_i} \mid p_1^{a_1} p_2^{a_2} \cdots p_{i-1}^{a_{i-1}} p_{i+1}^{a_{i+1}} \cdots p_n^{a_n}.$$

Since $b_i - a_i > 0$, we have that

$$p_i \mid p_1^{a_1}p_2^{a_2}\cdots p_{i-1}^{a_{i-1}}p_{i+1}^{a_{i+1}}\cdots p_n^{a_n},$$

from which $P_i \mid P_j$ for some $j \neq i$ by Corollary 1.15. Contradiction! Thus $a_i = b_i$, for $i = 1, 2, \ldots, n$, as required.
Example: $756 = 2^2 3^3 7$
Definition: Let $a, b \in \mathbb{Z}$ with $a, b > 0$. The least common multiple of a and b, denoted $[a, b]$J is the least positive integer m such that a/m and b/m.
Remark: Given two positive integers a, b, Consider the set

$$T := \{c : c \in \mathbb{Z}, c > 0, a \mid c, \text{ and } b \mid c\}.$$

We have $ab > 0$ and $ab \in T$, so $T \neq \phi$, and contains only positive integer, so the least common multiple always exists by the well-ordering Principle.

Example:

- Positive multiples of $6 : 6, 12, 18, 24, 30, 36, (42, 48, \ldots$

$$\text{of } 7 : 7, 14, 21, 28, 35, (42), 49, 54, \ldots$$

$[6, 7] = 42$.

- Positive multiples of $8 : 8, 16, 24, 32, 40, 48, 56, 64, \ldots$
  $[6, 8] = 24$
- The FTA can be used to compute gcds and lcms.

Proposition 1.17: Let $a, b \in \mathbb{Z}$ with $a, b > 1$. Write $a = p_1^{a_1}p_2^{a_2}\ldots p_n^{a_n}$ and $b = p_1^{b_1}p_2^{b_2}\ldots p_n^{b_n}$, where $p_1, \ldots, p_n$ are distinct primes and $a_1, a_2, \ldots, a_n, b_1, \ldots, b_n$ are non-negative integers (possibly zero). Then
$(a, b) = P_1^{\min\{a_1,b_1\}}P_2^{\min\{a_2,b_2\}}\ldots P_n^{\min\{a_n,b_n\}}$ and,
$[a, b] = P_1^{\max\{a_1,b_1\}}P_2^{\max\{a_2,b_2\}}\ldots P_n^{\max\{a_n,b_n\}}$.

Proof: This follows from the FTA and definitions of (a, b) and $[a, b]$.
(8)

Example: Find $(756, 2205)$ and $[756, 2205]$ using prime factorisations.

$$756 = 2^2 3^3 5^0 7^1$$
$$2205 = 2^0 3^2 5^1 7^2$$
$$(756, 2205) = 2^0 3^2 5^0 7 = 63$$
$$[756, 2205] = 2^2 3^3 5^1 7^2 = 26460$$

Remark:

- Prime factorisations most certainly do not make the Enclidean algorithm for computing gcds obsolete. Prime factorisations for large integers are hard to obtain!
  Lemma 1.18: Let $x, y \in \mathbb{R}$. Then

$$\max\{x, y\} + \min\{x, y\} = x + y.$$

Proof: If $x < y$, then $\max\{x, y\} = y$
and $\min\{x, y\} = x$, and the result follows. (9)
The cases for $x = y$ and $x > y$ are similar.

Theorem 1.19: Let $a, b \in \mathbb{Z}$ with $a, b > 0$.
Then $(a, b)[a, b] = ab$.
Proof: If $a, b > 1$, write

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$

and $\quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$)
with $p_1, \ldots p_n$ distinct primes and $a_1, \ldots, a_n$,
$b_1, \cdots$, bn won-negative integers. Then by
Proposition 1.17,
$(a, b)[a, b]$
$= P_1^{\min\{a_1,b_1\}} P_2^{\min\{a_2,b_2\}} \cdots P_n^{\min\{a_n,b_n\}}$
$\times P_1^{\max\{a_1,b_1\}} P_2^{\max\{a_2,b_2\}} \cdots P_n^{\max\{a_n,b_n\}}$
$= p^{\min\{a_1,b_1\}+\max\{a_1,b_1\}} \quad {}^{\min\{a_n,b_n\}+\max\{a_n,b_n\}}$
$\cdots P_n$
$a_n + b_n$
(by Lemma 1.18)
$= (p_1^{a_1} \ldots p_n^{a_n}) \left(p_1^{b_1} \ldots p_n^{b_n}\right)$
$= ab$, as required.
The cases $\begin{cases} a = 1 \text{ and } b > 1 \\ a > \pm \text{ and } b = 1 \\ a = b = 1 \end{cases}$
are left as an exercise.

- To find the 1 cm of two positive integers without using prime factorisation, compute their gcd using the Euclidean algorithm, and then use Theorem 1.19.
  Conollary 1.20: Let $a, b \in \mathbb{Z}$ with $a, b > 0$. Then $[a, b] = ab$ if and only if $(a, b) = 1$.
  Proof: Left as an exercise.
  Hint: Use Theorem 1.19.